

CSCE 679 DATA VISUALIZATION · MODULE 02 · LECTURE NOTES

The Channel Ranking

Measured accuracy, which decides most design questions.

MODULE OUTCOMES

State the ranking for quantitative and categorical data.

Explain why position dominates.

Explain separability and the channels that interfere.

Explain popout and its single-dimension limit.

Use the ranking as a prediction.

1 The ranking for quantitative data

Rank	Channel	Note
1	Position on a common scale	By a wide margin. Put your most important quantitative variable here (§4).
2	Position on unaligned scales	Still good; the comparison across separate panels is harder than within one (Module 06 §1).
3	Length	Which is why bars work, and why their axis must start at zero — see the callout.
4	Angle and slope	Slope judgement depends on the aspect ratio, which is a design parameter (Module 05 §4).
5	Area	Systematically underestimated — doubling the value looks like roughly 1.5 times.
6	Luminance and colour saturation	Ordered but imprecise; acceptable for a few distinguishable levels (Module 04 §3).
7	Colour hue	Not ordered at all — do not use it for quantity (Module 04 §2).

This table decides most chart questions before they become arguments — and it is **the single most useful artefact in the course**, which is why Module 01 §4 recommended memorising it. **Reproducing it from memory is worth doing**, because it is that load-bearing: §4's review, Module 04's colour rules, and Module 05's chart choices all reduce to it.

POSITION DOMINATES BECAUSE IT IS JUDGED AGAINST A SHARED REFERENCE

A position on a common axis can be compared against a gridline, against the axis itself, and against every other mark in the figure — so the judgement is relative and supported by multiple references rather than absolute.

Length comes next because it also has a zero reference — the bar's base — and **that is exactly why a truncated bar axis is a deception rather than a stylistic choice** (Module 10 §2): **it removes the reference the channel depends on**, so the length no longer means what the reader's visual system assumes.

And area is poor because there is no reference and the perceptual exponent is below one — people systematically underestimate area ratios, so **a bubble representing twice the value looks about one and a half times as large**, and the error is consistent rather than random.

Which means the ranking is not arbitrary: channels with a shared reference beat channels without one, and **that explains the order rather than merely stating it** — which makes it possible to reason about a channel the table does not list.

2 Categorical channels

- **Spatial region first** — small multiples, or separate panels, or distinct areas of one figure. **Effectively unlimited in the number of categories it holds, and the most accurately read**, which is why faceting is such a strong technique (Module 06 §1).
- **Then colour hue** — **which holds about five to seven categories before the reader has to consult the legend for each mark** (CSCE 671 Module 03 §2's working memory limit), after which it has stopped being a channel and become a lookup.
- **Then motion**, which is **extremely salient and holds very few categories** — and is intrusive enough that it should be reserved (CSCE 671 Module 02 §3).
- **Then shape**, which holds a moderate number and is **poor at small mark sizes**, where the shapes become indistinguishable — so it interacts with size (§3).
- **And note the inversion: hue is second for categories and last for quantity, which is the clearest demonstration that the channel ranking depends on the data type** (Module 03) — and is why **'is colour good'** is not a question with an answer until the data type is named.

3 Separability

SOME CHANNEL PAIRS ARE READ INDEPENDENTLY AND SOME INTERFERE

Position and colour hue are fully separable — a reader can judge a mark's position without its colour affecting the judgement, and vice versa — **which is why a colour-coded scatter plot works well** and is such a common and effective encoding.

But size and colour interfere: a small patch of colour is genuinely harder to identify than a large one, so **encoding one variable on size and another on hue degrades the reading of both** — and the degradation is worst exactly for the small marks, which biases the reading.

And width and height are not separable at all — the reader perceives the resulting area and aspect ratio rather than two independent values, which makes a two-variable rectangle encoding read as one variable plus a distraction.

So the number of variables you can usefully encode is bounded by separability rather than by the number of channels available — which is **the argument for small multiples and linked views** (Module 06) **over one dense figure**, and it is a perceptual argument rather than an aesthetic one.

- **Three or four encoded variables is a practical limit** for a single set of marks — and **beyond that, use faceting rather than adding more channels** (§2's first row, which is both accurate and unlimited).
- **Put the most important variable on position, always** — **which is the single decision the ranking most clearly settles** and the one most often got wrong by putting a decorative variable there instead.
- **Avoid size-plus-hue on the same marks**, and **never encode two variables on the two dimensions of a rectangle** — both of which follow directly from the callout.
- **And check for unintended encoding** — **if your marks differ in a channel for an incidental reason** (a default colour cycle, a size that varies with the label length), readers will attribute meaning to it.
- **Which is CSCE 671 Module 02 §1's spacing-wins result in another form: the visual system groups and orders whether you intended it to or not. Readers attribute meaning to any channel that varies, so an incidental difference in size or colour is a claim you did not mean to make** — and is worth auditing before a figure ships.

4 Using the ranking as a prediction

THE RANKING PREDICTS WHICH QUESTIONS A FIGURE WILL ANSWER WELL

Before showing a figure to anybody, you can predict which comparisons will be read accurately — the ones encoded on position, then on length — **and which will be guessed at**, the ones on area, luminance, or hue.

So a design review becomes a specific claim: 'the reader's main task is comparing these two values, and they are encoded on area, which is rank five' — which is a statement about reading accuracy rather than about taste, and is therefore arguable on evidence.

And it is directly testable: ask five readers to make the comparison and measure their accuracy and their time, which is **Project 1** and which **reliably confirms the ranking** — the effect sizes are large enough to see with five people.

Which is why this module is the one to internalise — it converts an aesthetic argument into a prediction that can be checked in an afternoon, and that conversion is what makes the rest of the course empirical rather than stylistic.

1. WHAT IS THE READER'S MAIN TASK?
a lookup, a comparison, a trend, a total, or a search for an extreme (Module 01 section 2)
2. WHICH VARIABLE DOES THAT TASK CONCERN?
3. WHICH CHANNEL IS IT ENCODED ON, AND WHAT RANK IS THAT?
4. IF IT IS NOT NEAR THE TOP, WHY NOT?
there are legitimate reasons -- the position channels are already in use, the figure has to serve several tasks at once, the data has more dimensions than there are good channels -- and the answer should be one of them rather than "it looked better".

THAT IS THE WHOLE REVIEW. It takes two minutes and it catches the large errors, which are nearly always a channel mismatch rather than a styling problem.

Two minutes, and it catches the large errors — which are nearly always a channel mismatch rather than a styling problem, and which no amount of attention to typography or colour palette will fix. **This four-question review is the course's most portable tool**: it requires only §1's table, it applies to any figure, and it produces a specific objection rather than a general dissatisfaction.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Cleveland & McGill — Graphical Perception (free)	§1's ranking in the original — the experiments that established it, and still the reference.
Munzner, chapter 5 (slides free)	§1 through §3 — the ranking, separability, and the popout material, organised.
Heer & Bostock — Crowdsourcing Graphical Perception (free)	§1 replicated at scale — which confirmed the original ranking and extended it.
Ware — Information Visualization, chapters 1 through 5	§1 and §3's perceptual basis — why the ranking has the shape it does. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Reproduce the quantitative ranking from memory**, and check it.
- 2 **Encode the same variable on position, length, and area**, and ask five people to estimate ratios from each.
- 3 **Measure the area underestimation** in your own readings.
- 4 **Truncate a bar axis** and ask people to compare two bars, then untruncate and ask again.
- 5 **Encode eight categories on hue** and time how long a lookup takes.
- 6 **Do the same with eight small multiples** and compare.
- 7 **Encode two variables on size and hue**, then separate them, and compare readings.
- 8 **Encode two variables on a rectangle's width and height**, and ask what people read.
- 9 **Find an incidental channel variation** in a figure you made.
- 10 **Run the four-question review** on five published figures.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give the quantitative ranking in order.
- 2 Why does position dominate, and why is length second?
- 3 Why is a truncated bar axis a perceptual problem rather than a stylistic one?
- 4 By how much is area misread, and in which direction?
- 5 Give the categorical ranking and the capacity of hue.
- 6 What is the inversion between the two rankings, and what does it show?

- 7 Name a separable pair and two interfering pairs.
- 8 What bounds the number of variables you can encode?
- 9 Why do readers attribute meaning to incidental variation?
- 10 Give the four-question design review.